

Pentium Processor Architecture & Superscalar Operation MPCA Module 5

> **Definition:** The **Intel Pentium Processor** is a 32-bit superscalar CISC microprocessor capable of executing **two instructions per clock cycle** using two parallel 5-stage integer pipelines (**U-Pipe** and **V-Pipe**), an integrated on-chip Floating-Point Unit (FPU), and **Dynamic Branch Prediction Logic**.

1. Detailed Technical Explanation

1. Pentium Architecture Block Diagram

...

PENTIUM PROCESSOR ARCHITECTURE BLOCK DIAGRAM

2. Superscalar Integer Pipelines (U and V Pipes)

The Pentium can issue two integer instructions simultaneously in parallel if they meet **Instruction Pairing Rules**:

The 5 Pipeline Stages:

...

1. PF (Pipeline Fill)

3. Dynamic Branch Prediction & Branch Target Buffer (BTB)

- **Branch Target Buffer (BTB):** An on-chip cache storing branch instructions, target jump addresses, and branch history bits.

- **2**

4. Must-Write Points for Exams

- Pentium has a **64-bit external data bus** and **32-bit address bus**.

- It fe

5. Quick Recall Flow

Pentiu
Predic